FYP-2 Mid Evaluation — 5-Minute Presentation Script

S26-017-R-GASR · FAST NUCES Islamabad · Supervisor: Dr Akhtar Jamil

Slides 1–7 are the talk. Slides 8–12 are the demo. Slides 13–23 are held for questions.

Member	Slides	Words	Time	Section
Ahmad Rashad 22i-1175	1 – 3	247	1:46 – 1:59	Goals and objectives · faculty feedback addressed
Shehroz 22i-0771	4 – 6	295	2:06 – 2:22	Progress since FYP-1: methodology, results, literature
Humair Khan 22i-2623	7	88	0:38 – 0:42	Remaining work, then drives the demo
Total		**630**	**4:30 – 5:02**	Counted from the spoken lines only. Fast end = 140 wpm, slow end = 125 wpm. Either way you are inside five minutes.

> **How to use this.** These are the words, not notes to expand on. The deck carries the detail — do not read the slides out. Time yourself: if a section runs long, you are adding sentences that are not here. **Do not stop for questions during slides 1–7**; the committee's five minutes are uninterrupted.

Ahmad Rashad — Slides 1 to 3 · about 1 : 50

Slide 1 — Title · 28 words · 12 s

Good morning. I'm Ahmad Rashad, with Shehroz and Humair. Our supervisor is Dr Akhtar Jamil. Project S26-017 — a research FYP on 4D surgical reconstruction and deformation prediction.

Slide 2 — What the project is for · 114 words · 49 s

A surgeon operating through one laparoscope sees a flat image of the present moment. Our goal is a system that reconstructs that scene in 3D and anticipates how the tissue deforms next — one camera, no depth sensor, no clinical annotation.

Point along the four cards. Say every status — the panel is marking progress.

Four objectives. Reconstruct in 3D from monocular video: complete, that was FYP-1. Learn a self-supervised model of tissue deformation: complete, Module 5. Predict future geometry with calibrated confidence: implemented, Module 6. Evaluate against baselines and the literature: in progress, Modules 7 and 8.

Pause. Read the question slowly — it is the sentence they should remember.

So the research question is this. Can future 3D tissue geometry be predicted from monocular surgical video

alone — and can the system say how much to trust each prediction?

Slide 3 — Faculty feedback from FYP-1 · 105 words · 45 s

Four rows. Say the feedback, say what was built. The files are on the slide — do not read them aloud.

We had four pieces of feedback after FYP-1. All four are implemented.

Implement a base paper for comparative analysis — we implemented two. PointRNN, the closest published method to ours, and Luo and Hu's point-cloud diffusion.

Implement state-of-the-art papers — eleven methods on one protocol. Five classical, five learned, plus ours, all scored on identical windows.

Detailed hyperparameter and optimisation analysis — Module 8, four studies.

And write a paper in IEEE or Springer format — a six-page IEEE manuscript with 22 references, in the repository.

All four are written. Two still need their compute runs, and that's item one on our remaining-work slide.

Step back from the laptop.

Shehroz.

Shehroz — Slides 4 to 6 · about 2 : 10

Slide 4 — The methodology, finalised · 95 words · 41 s

The methodology is now fixed. Given the last four frames of 512 tracked tissue points, predict their displacement one, two or three frames ahead — with no clinical labels at any stage.

It's a conditional diffusion model, 16.3 million parameters, v-prediction on a cosine schedule.

The middle card. This is the one decision worth naming.

One design decision matters most. It doesn't predict the displacement — it predicts the correction to constant-velocity extrapolation. So a model that learns nothing still beats the naive baseline by 41%.

Ten videos, 7,371 windows, self-supervised throughout. The method is written up in full in Section III of the manuscript.

Slide 5 — Metrics chosen, and what they measured · 118 words · 51 s

Lead with why the metrics — that is what is being marked.

Metrics, and why these ones. Chamfer Distance is our primary metric because it's correspondence-free. But it cannot separate a model that learned nothing from one that learned the right direction and the wrong distance — both score the same. So we added a directional cosine, which separates exactly those two, and a spread–error correlation to test whether the uncertainty is calibrated.

What they measured. On accuracy we're not yet ahead — and constant velocity beats us on all ten videos. On direction, cosine plus 0.57; a model that learned nothing reads zero, so our failure is magnitude, not direction. And uncertainty correlates with real error at 0.69.

We also tested and eliminated drift and dropped frames as explanations.

Slide 6 — How we compare with published work · 82 words · 35 s

Say the caveat yourself. Do not let them find it.

Against the literature. For reconstruction our base papers are LerPlane, EndoGaussian and Deform3DGS, at 29.46, 30.25 and 30.48 dB. We report 30.10 — but those use stereo on different sequences, so it's indicative, not controlled. What it does establish is that we're within 0.4 dB of the stereo state of the art using a single camera.

For prediction: PointRNN as our learned baseline, Luo and Hu as our diffusion base paper. The full eleven-method comparison is implemented; the run is next.

Step back.

Humair.

Humair Khan — Slide 7 · about 0 : 40, then the demonstration

Slide 7 — Remaining work until the final evaluation · 88 words · 38 s

Give each item a status, not a hope.

Five things remain, in the order they have to happen.

One: run the eleven-method comparison and the tuning studies. Both are written and tested — what's left is compute, not code.

Two: evaluate the two corrections we've already implemented. Calibration needs no retraining.

Three: move to leave-one-video-out evaluation.

Four: fold the results into the manuscript — its tables are generated from result files, so finished runs populate it directly.

Five: submit, and grow the corpus beyond ten recordings.

Go straight into the demo. Do not pause for questions.

And with that, let me show you the system running.

The Demonstration — Humair drives, slides 8 to 12

No script here, because this is a live system and it will not run to a script. These are the lines to have ready.

Slide 8 — while the notebook loads

Everything from here is live — nothing is pre-rendered. We load the checkpoint, verify the evaluation data, predict a held-out moment, render it as an actual video frame, compare it with the real frame, and then forecast past the end of the recording.

Slide 9 — while the render cell runs

A prediction moves 512 points; a picture needs all 58,000 Gaussians. The bridge is a distance-weighted interpolation, weighted so a tracked point gets its own predicted motion exactly.

Two things about what you're seeing. Motion at one to two pixels is invisible, so the arrows are magnified — and that factor is printed on every panel and never applied to any number. And the forecast runs twelve frames past the end of the recording: the real-video panel stops, the predicted panel keeps going. That's the only output that shows prediction rather than describing it.

Slides 10 to 12 — describing the results

Agree beforehand whether you or Shehroz narrates these. One of you, not both.

Slide 10, the key finding: the model knows which way tissue moves, not how far. Slide 11: when the five sampled futures disagree, the prediction really is worse — correlation 0.69. A system that is confidently wrong is unusable in theatre; one that says "don't trust me here" is usable even when imperfect. Slide 12: the full comparison against both baselines at all three horizons.

> **Demo checklist — do this the day before, not on the day.**
>
> - Runtime connected, GPU confirmed, checkpoint verified loadable.
> - Run the whole demo path end to end twice. Time the slowest cell and know what you'll say over it.
> - Deck, notebook, checkpoint and outputs on **one laptop** — the committee requires it.
> - Fallback: if the runtime dies, show the already-executed notebook and say plainly that it is not re-executing. Do not pretend.
> - Screen resolution set so notebook text is readable from the back of the room.

Questions

Topic	Answers · slides
Goals, scope, the model, diffusion, training, overfitting, architecture, the FYP-1 feedback	**Ahmad** · 14–16, 19
Methodology, metrics, baselines, quality tests, the comparison, the cosine finding, uncertainty	**Shehroz** · 17–18, 20
The demo, rendering, the forecast, reconstruction quality, the dataset	**Humair** · 9, 21

The one answer to agree on. "Your model doesn't beat a two-line baseline — why should this pass?" One person answers: *"Because we can tell you exactly why it doesn't. We measured that it learns motion direction at 0.57 and fails only on magnitude, we eliminated drift and dropped frames as explanations, and we implemented the two corrections that follow. We also produced a calibrated uncertainty estimate no baseline can give you. For a research project, a measured negative with an identified cause is worth more than an unexplained positive."*